

Solution Architecture

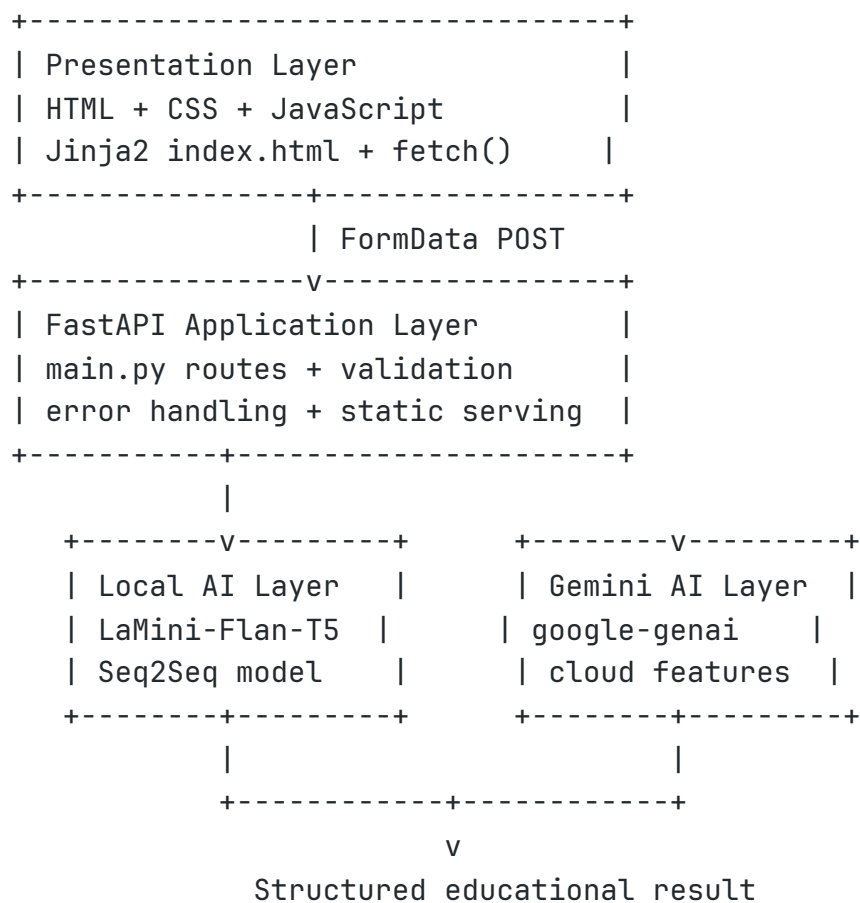
Date: 30 September 2026

Team ID: SWTID-2026-3656

Project Name: EduGenie – Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

High-Level Architecture



Component Table

Component	Role	Technology
Presentation	Collect input and display results	HTML/CSS/JavaScript/Jinja2
FastAPI routes	Expose five endpoints	FastAPI + Uvicorn
Explanation module	Generate local concept explanations	Transformers + PyTorch
Gemini helper	Shared cloud text generation	google-genai
Quiz validator	Parse and enforce five-question JSON	Python json/regex
Configuration	Keep key outside source	.env + python-dotenv